

The Macroeconomic Effects of Migration Structures

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1 Graphs

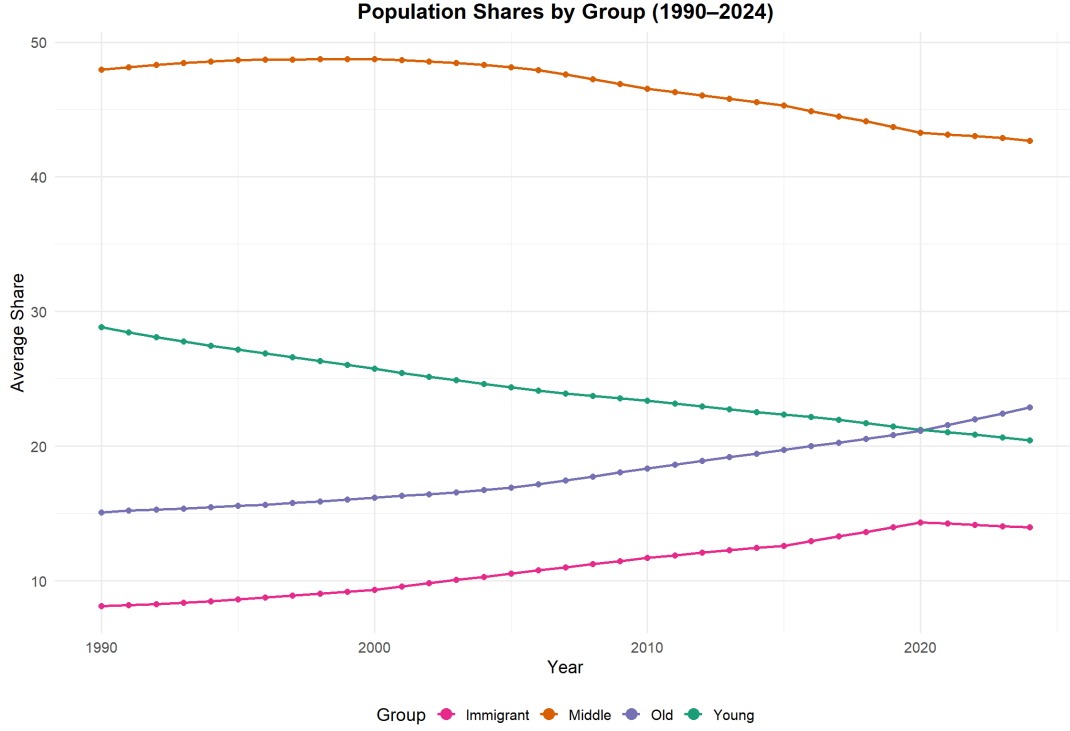


Fig. 1

Table 1: Effects of Population Variables on Macro Variables

Variable	young	mid	old	pop _g	immigrant
cpi	0.634 [0.334, 0.928]	-0.257 [-0.569, 0.053]	-0.027 [-0.309, 0.265]	0.340 [-0.877, 1.634]	-0.349 [-0.587, -0.115]
gdp _p	0.033 [-0.006, 0.071]	0.083 [0.039, 0.127]	-0.135 [-0.175, -0.097]	0.033 [-0.183, 0.268]	0.019 [-0.010, 0.049]
interestrate	0.686 [0.474, 0.901]	-0.180 [-0.396, 0.030]	-0.137 [-0.327, 0.059]	0.011 [-0.689, 0.708]	-0.369 [-0.543, -0.202]
inv	-0.271 [-0.468, -0.087]	0.398 [0.191, 0.611]	-0.306 [-0.490, -0.123]	0.037 [-0.846, 0.881]	0.179 [0.039, 0.332]
savingrate	-0.479 [-0.882, -0.117]	0.390 [-0.012, 0.827]	-0.202 [-0.586, 0.177]	-0.615 [-2.801, 1.172]	0.291 [0.014, 0.603]
unem	0.122 [-0.110, 0.376]	0.011 [-0.268, 0.278]	-0.013 [-0.258, 0.234]	0.535 [-0.873, 2.192]	-0.120 [-0.319, 0.063]

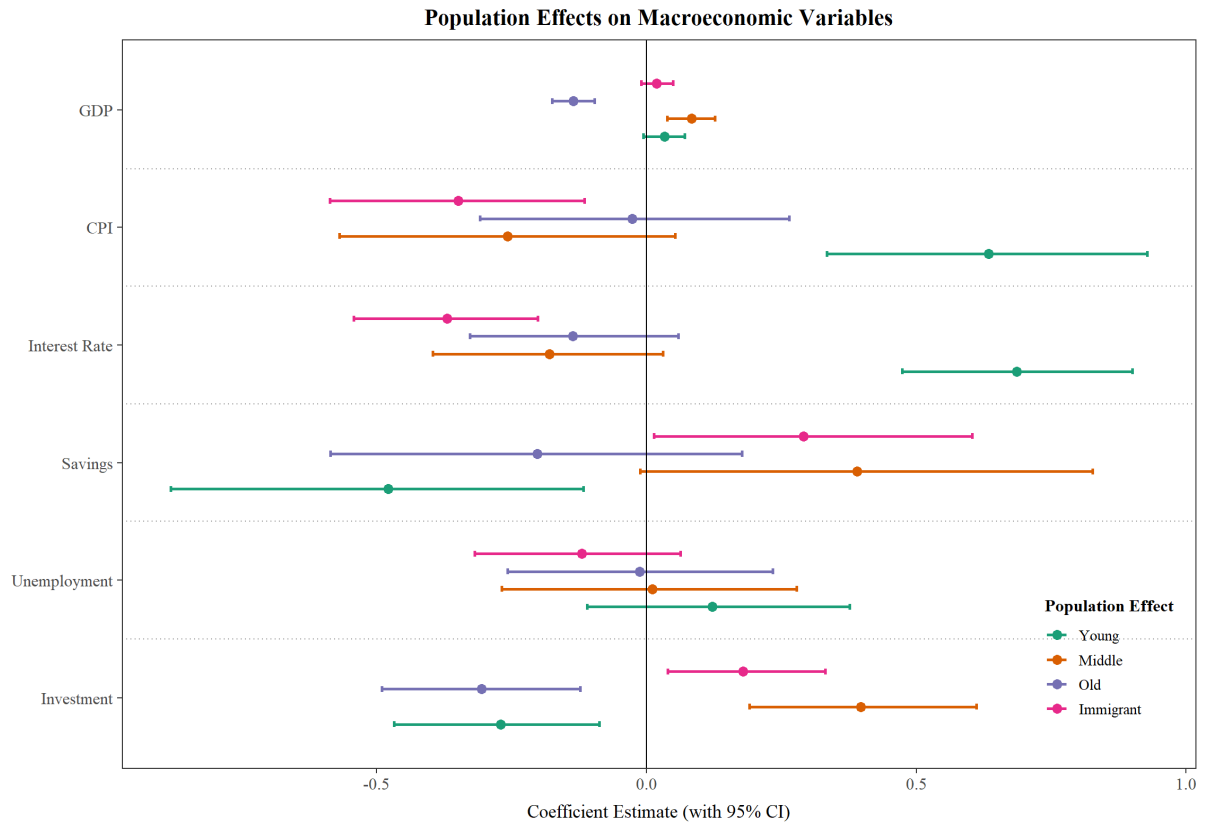


Fig. 2

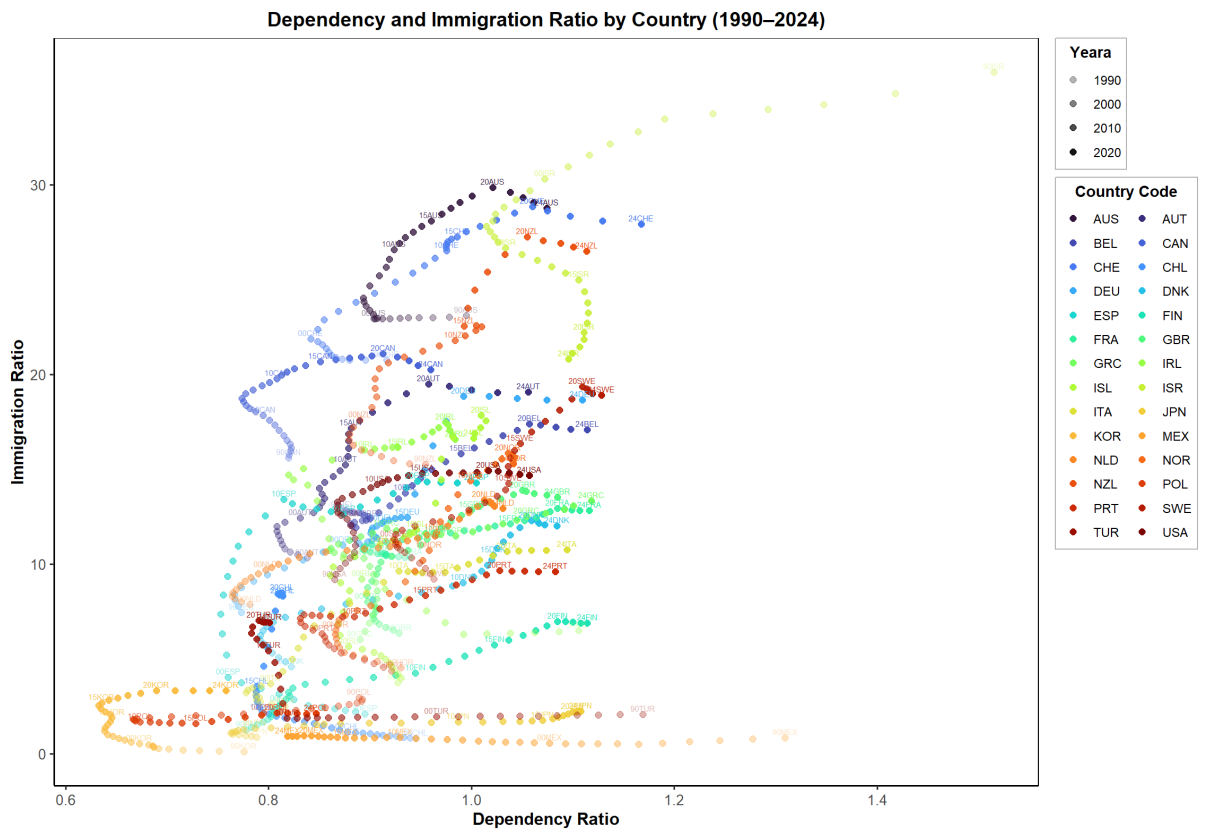


Fig. 3

A scatter plot showing the relationship between the Immigration Ratio (Y-axis, 0 to 35) and the Dependency Ratio (X-axis, 0.6 to 1.5). The plot displays data points for 150 countries, categorized into three clusters based on their dependency and immigration ratios. The clusters are represented by different colors and markers:

- Cluster 1 (Green dots):** Countries with low dependency ratios (around 0.7-0.8) and low immigration ratios (around 5-10). Examples include KOR, POL, ESP, ITA, and FIN.
- Cluster 2 (Orange dots):** Countries with moderate dependency ratios (around 0.8-1.0) and moderate immigration ratios (around 5-15). Examples include AUS, CAN, CHE, ISR, NZL, and DEU.
- Cluster 3 (Blue dots):** Countries with high dependency ratios (around 1.0-1.5) and high immigration ratios (around 15-35). Examples include AUS, CAN, CHE, ISR, NZL, and DEU.

Three cluster centers are marked with red dots and labeled with their coordinates (Dependency Ratio, Immigration Ratio):

- Cluster 1 center: (0.81, 5.52)
- Cluster 2 center: (0.97, 10.36)
- Cluster 3 center: (1.00, 23.48)

Fig. 4

The forest plot displays the estimated coefficients and 90% confidence intervals for three clusters across six economic variables. The x-axis represents the Estimated Coefficient (90% CI), ranging from -2 to 1. The y-axis lists the variables: GDP, CPI, Interest Rate, Savings, Unemployment, and Investment. The legend indicates three clusters: Cluster 1 (green), Cluster 2 (orange), and Cluster 3 (blue).

Variable	Cluster	Estimated Coefficient (90% CI)
GDP	Cluster 1	-0.05 (-0.15, 0.05)
	Cluster 2	0.10 (0.00, 0.20)
	Cluster 3	-0.05 (-0.15, 0.05)
CPI	Cluster 1	-0.05 (-0.15, 0.05)
	Cluster 2	-0.50 (-1.00, 0.00)
	Cluster 3	-0.05 (-0.15, 0.05)
Interest Rate	Cluster 1	-0.20 (-0.30, -0.10)
	Cluster 2	-0.40 (-0.50, -0.30)
	Cluster 3	0.05 (-0.05, 0.15)
Savings	Cluster 1	-0.10 (-0.20, 0.00)
	Cluster 2	-0.10 (-0.20, 0.00)
	Cluster 3	-0.80 (-1.00, -0.60)
Unemployment	Cluster 1	-0.05 (-0.15, 0.05)
	Cluster 2	0.20 (0.10, 0.30)
	Cluster 3	-0.10 (-0.20, 0.00)
Investment	Cluster 1	-0.10 (-0.20, 0.00)
	Cluster 2	-0.10 (-0.20, 0.00)
	Cluster 3	-0.10 (-0.20, 0.00)

Cluster	Dependency	Immigration
Cluster 1	Low	Low
Cluster 2	High	Low
Cluster 3	High	High

Fig. 5

2 Research Question

How do immigrants affect economic performance, and what factors drive the variation in their impact?

3 Hypotheses

- H1.** Countries with greater *immigration urgency* will experience a positive long-run effect of immigration on the economy
- H2.** Countries with greater *Immigrant inclusiveness* will experience a positive long-run effect of immigration on the economy (to be done)
- H3.** Diversity in immigrants' countries of origin will make a positive long-run effect of immigration on the economy (to be done)

4 Model Specification

4.1 Variables

We begin by defining the PVARX model, treating population structure as an exogenous variable, following Aksoy et al., 2019 with bayesian inference:

$$Y_{it} = (g_{it}, \pi_{it}, i_{it}, u_{it}, s_{it}, iv_{it})',$$

comprises six macroeconomic variables. The macro variables are real GDP growth g_{it} , the short-term interest rate i_{it} , CPI π_{it} , the investment-to-GDP ratio i_{it} , the personal savings-to-GDP ratio s_{it} and unemployment rate u_{it} .

The exogenous control vector is

$$B_{it} = (na_{1,it}, na_{2,it}, na_{3,it}, pop_gr_{it})',$$

We divide the native population into three age-groups using cut-offs at ages 19 and 59: $Na_{1,it}$, $Na_{2,it}$, and $Na_{3,it}$ denote the shares of the native population aged below 19, between 20 and 59, and 60 or above, respectively. Then we define the native population shares

$$\{Na_{1,it}, Na_{2,it}, Na_{3,it}\}$$

are transformed to :

$$na_{1,it} = Na_{1,it} - imm_{it}, \quad (1)$$

$$na_{2,it} = Na_{2,it} - imm_{it}. \quad (2)$$

$$na_{3,it} = Na_{3,it} - imm_{it}. \quad (3)$$

This ensures no perfect collinearity.

4.2 Model Specification

To address overfitting in the moderately sized panels, we impose the following hierarchical prior structure in regression model:

$$Y_t = \alpha + A Y_{t-1} + B X_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \Sigma), \quad (4)$$

where

- α is an 6×1 intercept vector.
- A is the 6×6 autoregressive coefficient matrix at lag 1.
- B is the 6×4 matrix of exogenous (population-effect) coefficients.
- Σ is the 6×6 residual covariance matrix.

Minnesota-Gamma Prior

$$A_{ij} \sim \mathcal{N}(\mu_{ij}, \sigma_{A,ij}^2), \quad \sigma_{A,ij} = \lambda_1 |i = j| \times \begin{cases} 1 & \text{if } i = j, \text{ (own lag)} \\ \lambda_2 & \text{if } i \neq j \text{ (cross lag)} \end{cases}, \quad (5)$$

$$B_{k,i} \sim \mathcal{N}(0, \lambda_1 \lambda_3), \quad (6)$$

$$\lambda_1, \lambda_2, \lambda_3 \sim \Gamma(0.1, 0.1), \quad (7)$$

where $\mu_{ij} = 1$ if $i = j$ (own-lag coefficient) and zero otherwise. The hyperparameters $(\lambda_1, \lambda_2, \lambda_3)$ control overall shrinkage, cross-equation shrinkage, and exogenous-variable shrinkage, respectively, and themselves are given weakly informative Gamma priors.

Error and Initial Conditions

$$\Sigma \sim \mathcal{IW}(\nu_0, \Psi_0),$$

with degrees of freedom $\nu_0 = M + 1$ and scale matrix $\Psi_0 = (0.02)^2 I_M$.

All posterior inference is carried out via Hamiltonian Monte Carlo using the `rstan` package, with 4 chains of 2000 iterations (1000 warmup) and an adapt constant of 0.9.

4.3 Estimated Shrinkage Hyperparameters

Table 2 reports the posterior means and 90% credible intervals for the three global shrinkage parameters. The estimated values are in line with those found in Bayesian VAR studies and confirm that our choice of $\text{Gamma}(0.1, 0.1)$ priors produces sensible amounts of shrinkage.

Table 2: Posterior Estimates of Shrinkage Hyperparameters

Parameter	Posterior Mean	min	max
λ_1	0.4370	0.2451	0.8137
λ_2	0.1712	0.0783	0.3042
λ_3	0.1776	0.0770	0.3295

We use these values in our subgroup models as a fixed parameter.

4.4 Long-Run Effects

Once the posterior draws of the autoregressive matrix A and the exogenous-coefficient matrix B are obtained, we compute the long-run multiplier matrix Θ as

$$Y_t = \alpha + A Y_t + B X_t \implies Y_t = (I - A)^{-1}(\alpha + B X_t) = (I - A)^{-1} \alpha + (I - A)^{-1} B X_t$$

$$\Theta = (I - A)^{-1} B$$

For the immigrant cohort, the long-run effect of a one-percentage-point increase in the immigrant (population cohort) share on the macroeconomic outcome. We derive these effects by computing for each posterior draw $(A^{(s)}, B^{(s)})$ the matrix

$$\Theta^{(s)} = (I - A^{(s)})^{-1} B^{(s)},$$

and then report the posterior mean and credible interval for each element Θ_{ij} .

Computation and Inference

- Draws $\{A^{(s)}, B^{(s)}\}_{s=1}^S$ are obtained via HMC in `rstan`.
- For each draw s , form $\Theta^{(s)} = (I - A^{(s)})^{-1} B^{(s)}$.
- Report $\mathbb{E}[\Theta_{ij} \mid \text{data}]$ and 90% credible interval $[\Theta_{ij}^{(0.025)}, \Theta_{ij}^{(0.975)}]$.

Table 3: Data Source and Notes

Category	Variable	Source	Notes
Macroeconomic Indicators	GDP growth	IMF WEO, ₂₅	Domestic currency, constance price
	Real GDP growth	IMF WEO, ₂₅	—
	CPI	IMF WEO, ₂₅	—
	Short-term interest rate	FRED	3months or 90days interbank rate
	Investment-to-GDP ratio	IMF WEO, ₂₅	Private
	Savings-to-GDP ratio	IMF WEO, ₂₅	Private
	Unemployment rate	IMF WEO , ₂₅	—
Demographic Indicators	Immigrant ratio	UNPD IMS, ₂₆	Five-year linear interpolated
	Native age structure	UNPD WPP, ₂₅ July 1	
Immigrant Indicators	some indicators(not decided)	MIPeX	—
	immigrant unemp rate	ILO stat	—

References

- Aksoy, Y., Basso, H. S., Smith, R. P., and Grasl, T. (2019). Demographic structure and macroeconomic trends. *American Economic Journal: Macroeconomics*, 11(1):193–222.